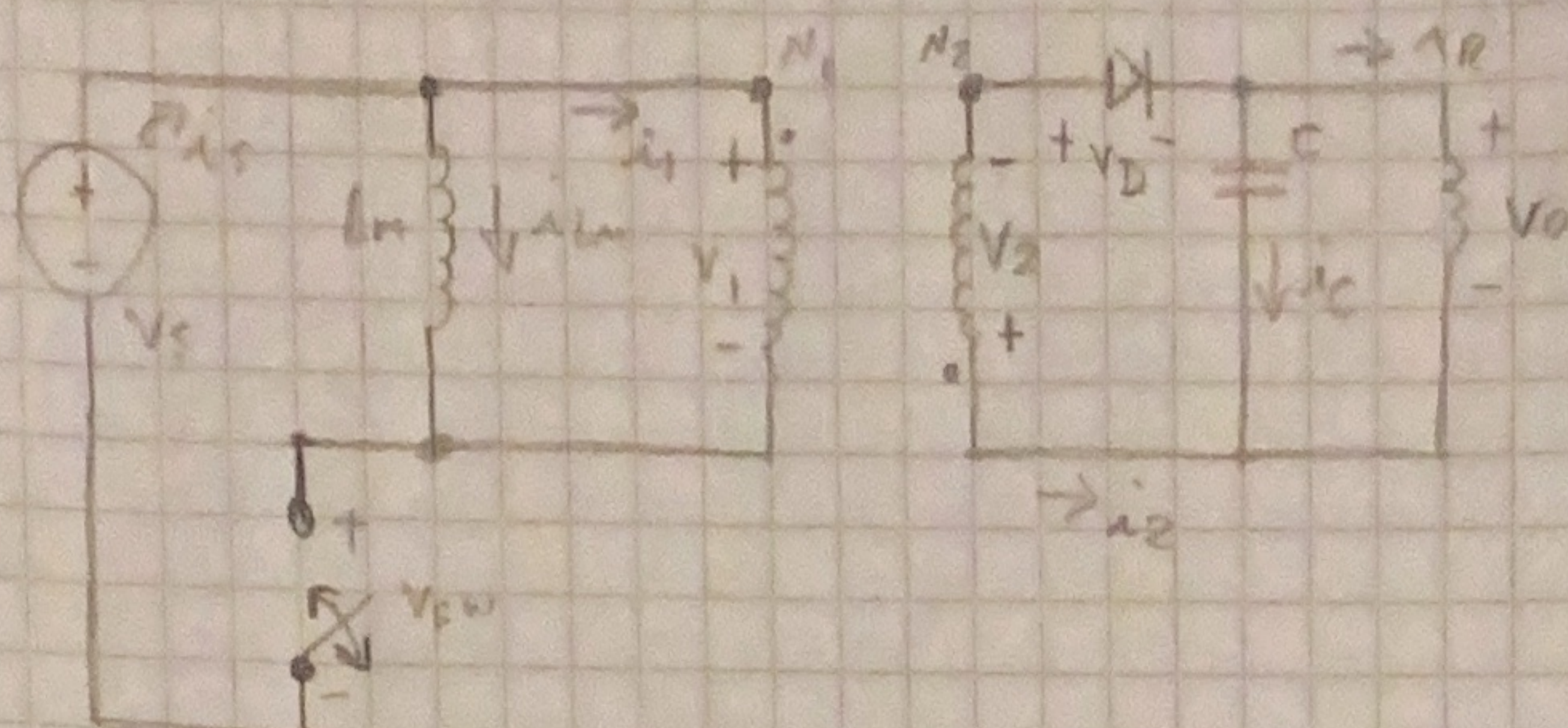




DATOS:

- $V_o = 5V$
- $0,5A \leq i_O \leq 1,3A$
- $V_s = [270, 330]V$

1. LA TENSION MEDIA EN EL INDUCTOR DEBE SER CERO, ENTONCES:

$$V_s \cdot \frac{I_{on}}{T_s} = \frac{N_1}{N_2} \cdot V_o (T_s - I_{on}) \rightarrow V_s \cdot \frac{I_{on}}{T_s} = \frac{N_1}{N_2} \cdot V_o \left(1 - \frac{I_{on}}{T_s}\right) \rightarrow$$

$$V_s \cdot D = \frac{N_1}{N_2} \cdot V_o (1 - D) \rightarrow \frac{N_2}{N_1} = \frac{V_o}{V_s} \cdot \left(\frac{1 - D}{D}\right)$$

$$\text{Si } D = 0,5 \text{ y } V_s = 300V \rightarrow \frac{N_2}{N_1} = 0,16$$

$$2. \frac{N_2}{N_1} \cdot V_s \cdot D = V_o - V_o \cdot D \rightarrow \left(V_s \cdot \frac{N_2}{N_1} + V_o\right) \cdot D = V_o \rightarrow$$

$$D = \frac{V_o}{V_s \cdot \frac{N_2}{N_1} + V_o} \xrightarrow{V_s = 270V} D = 0,103 \text{ MAXIMO}$$

$$\xrightarrow{V_s = 330V} D = 0,08 \text{ MINIMO}$$

3. SABIENDO QUE SE CUMPLE:

$$\begin{cases} i_s = \frac{N_2}{N_1} \cdot \frac{D}{1-D} \cdot i_o \\ i_s = D \cdot i_{LM} \end{cases} \text{ IGUALO: } D \cdot i_{LM} = \frac{N_2}{N_1} \cdot \frac{D}{1-D} \cdot i_o$$

$$\therefore i_{LM} = \frac{N_2}{N_1} \cdot \frac{i_o}{1-D} \xrightarrow{i_o = 1A, D_{min}} i_{LM} = 0,08A$$

$$\xrightarrow{i_o = 1,3A, D_{max}} i_{LM} = 0,23A$$

4. USANDO $i_s = D \cdot i_{LM}$

$$\text{Si } i_{LM} \text{ ES MAX } (D_{max}): i_s = 0,23A \cdot 0,103 = 0,023A$$

$$\text{Si } i_{LM} \text{ ES MIN } (D_{min}): i_s = 6,4mA$$

5. con $i_{Lmin} = 0,08 \text{ A}$, $\Delta i_L = i_{Lmax} - i_{Lmin} = 0,23 \text{ A} - 0,08 \text{ A}$

$$\Delta i_L = 0,15 \text{ A}$$

. PICO MAX DADO POR:

$$I_D = I_{Lmax} + \Delta I_L / 2$$

$$I_D = 0,3 \text{ A}$$

HOJA 2 ANEXA